have no effect on whether it is orthogonal to |*r*>, though in general the result will no longer be normalized (have unit length). But if we choose $z = e^{i\theta}$ (where θ can be any real number), then there will be no effect on the normalization because $e^{i\theta}$ has unit magnitude. In other words, $\alpha_u^* \alpha_u + \alpha_d^* \alpha_d$ will remain equal to 1. Since a number of the form $z = e^{i\theta}$ is called a phase-factor, the ambiguity is called the phase ambiguity. Later, we will find out that no measurable quantity is sensitive to the overall phase-factor, and therefore we can ignore it when specifying states.

2.4 ALONG THE *Y* AXIS

Finally, this brings us to |*i*> and |*o*>, the vectors representing spins oriented along the *y* axis. Let's look at the conditions they need to satisfy. First,

$$\langle i|o\rangle = 0. \qquad (2.7)$$

This condition states that *in* and *out* are represented by orthogonal vectors in the same way that *up* and *down* are. Physically, this means that if the spin is *in*, it is definitely not *out*.

There are additional restrictions on the vectors |*i*> and |*o*>. Using the relationships expressed in Eqs. 2.1 and 2.2, and the statistical results of our experiments, we can write the following: